# Open problems:
# Unitary evolution as infinitesimal projective processes

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

## From the brief

1. **Thermodynamic analogy.** The brief compares the result with the way "reversible quasi-static evolution in thermodynamics can be understood as a limit of equilibration processes". No precise statement linking the two limits is given. (Brief, Sec. 1; Sec. 6: "a reversible quasi-static limit of projective processes".)

2. **Resolvent of an unbounded generator.** The brief states that "direct use of the resolvent of an unbounded generator still gives a completely positive trace-preserving map and the resolvent exponential formula", and that the spectral truncations are needed only for states outside the domain of $H$. Neither claim is proved, and whether the untruncated $(I - \tau\mathcal{L})^{-1}$ gives perfect prediction on the domain of $H$ is left open. Not formalized. (Brief, Sec. 5, "Unbounded generators"; Log, Sec. 1.1, "Differences", last item.)

## From the formalization

Beyond finite dimension the formalization has a part in any dimension, which covers unbounded $H$ in DR-UNIT ⇒ DR-PROJ and derives the evolution structure `IsEvolution` from DR-PROJ up to the item "Durations with irrational ratio" below (`drUnit_iff_drProj_and_map_zero_and_map_add`). It leaves open the item "Resolvent of an unbounded generator" above and the items "Complete positivity", "The generator of the cutoff family" and "General cutoff levels" below (the first is open in finite dimension too).

1. [BLOCKS VERIFICATION: BRIEF ISSUE] **Durations with irrational ratio.** DR-PROJ defines $\Phi_t$ only as $\lim_n (\Psi_{t/n})^n$. For $t/s$ irrational a common step $\tau$ gives blocks $\Psi_\tau^k$ that are not of this form, so "common refinements" and continuity at zero do not prove $\Phi_{t+s} = \Phi_t \circ \Phi_s$; eq. (26) would have to hold uniformly along the trajectory. The remaining interval itself is harmless. A fix: in finite dimension a telescoping estimate gives $\Phi_t = e^{tL}$ directly, which is formalized (`IsProjectiveLimit.exists_eq_exp`); in infinite dimension Chernoff's product formula does, if the closure of $L$ generates a contraction semigroup, which is open. In infinite dimension this is the one clause of `IsEvolution` that is assumed; the others, including composition for durations of rational ratio (eq. (29)), are derived (`IsProjectiveLimit.isEvolution`).

$\Phi_0 = I$ is assumed in both dimensions, since the formalization defines $\Phi_t$ only for $t > 0$; the brief obtains it from $\Psi_0 = I$.

The gap may be real in infinite dimension. A sketch, not checked in Lean: let $H_1 \neq H_2$ be two self-adjoint extensions of a symmetric operator $H_0$ with deficiency indices $(1, 1)$, and let $\Psi_\tau$ be the cutoff resolvent of $H_1$ for rational $\tau$ and of $H_2$ for irrational $\tau$. Since $t/n$ is rational exactly when $t$ is, limits, prediction and reconstruction hold for each $t$, and $\Phi_t$ is conjugation by $e^{-itH_1}$ or $e^{-itH_2}$. If both families have the derivative $[H_0, \rho]/(i\hbar)$ on the density operators built from vectors of $\mathcal{D}(H_0)$, which are dense, then DR-PROJ holds while DR-UNIT would force $H_1 = H_2 + c$. (Brief, Sec. 4, after eq. (29); Log, "Problems found".)

2. [BLOCKS VERIFICATION: BRIEF ISSUE] **What perfect reconstruction means.** DR-PROJ only asks that the probabilities of the reconstructed sequences tend to one. The formalization instead assumes a second projective limit $\Phi'_t$ with $\Phi'_t(\Phi_t P) = P$ and $\Phi_t(\Phi'_t P) = P$ on pure states. It is not settled whether the brief's weaker wording already makes $\Phi_t$ a bijection of the pure states, or what the right formulation is. (Brief, Sec. 4: "Perfect reconstruction gives the converse correspondence"; Log, "Differences" and "Problems found".)

3. **Complete positivity.** Proposition 2 says that $\Psi_\tau$ is completely positive, as a mixture of unitary channels by eq. (7). Only positivity and trace preservation are formalized, in either dimension. (Log, "Differences"; Sec. 1.1, "Differences", last item.)

4. **The generator of the cutoff family.** For unbounded $H$ the brief states that the family (23) is infinitesimal with generator $\mathcal{L}(\rho) = [H, \rho]/(i\hbar)$. The formalization only shows that $(\Psi_\tau(\rho) - \rho)/\tau$ converges on the core; it does not identify the limit. (Brief, Sec. 4, after eq. (23); Log, Sec. 1.1, "Differences", item on eq. (26).)

5. **General cutoff levels.** DR-UNIT $\Rightarrow$ DR-PROJ for unbounded $H$ is formalized for the single choice $M_\tau = \tau^{-1/4}$. The brief's argument for every cutoff satisfying eq. (22) is not formalized. (Brief, eq. (22); Log, Sec. 1.1, "Differences".)

6. **The qubit picture.** $\Psi_\tau = (I + \mathcal{C}_\tau)/2$ with the Cayley transform $\mathcal{C}_\tau = (I + \tau\mathcal{L})(I - \tau\mathcal{L})^{-1}$, its orthogonality for the Hilbert–Schmidt inner product, and the midpoint picture on the Bloch ball are not formalized. (Brief, Sec. 5, "The qubit picture"; Log, "Problems found", last item.)

7. **The finite process is not unitary.** The claims that $\Psi_\tau$ is generally irreversible, maps pure ensembles to mixed ones, and that $I - \tau\mathcal{L}$ is not generally positive are not formalized. (Brief, Sec. 5, "The finite process is not unitary"; Log, "Problems found", last item.)